Innovation and Reconstruction of Early Childhood Education Models Driven by Artificial Intelligence Technology

Abstract

The rapid advancement of artificial intelligence (AI) is fundamentally reshaping global educational ecosystems, with early childhood education—the cornerstone of lifelong learning—undergoing unprecedented structural transformations. This study employs sociotechnical theory and educational ecology frameworks to analyze AI's innovative applications in preschool settings, revealing its profound impacts on pedagogical restructuring, teacher-child relationship evolution, and value system shifts. Key findings demonstrate that intelligent educational robots, virtual reality (VR) learning environments, and adaptive learning systems transcend traditional spatiotemporal boundaries, enabling data-driven personalized education. However, challenges such as algorithmic bias exacerbating educational inequity, privacy risks in child data management, and emotional interaction deficits demand urgent resolution. The proposed "technology-education-ethics" collaborative governance framework emphasizes child-centered values, advocating for legislative safeguards, teacher competency enhancement, and multi-stakeholder engagement to ensure sustainable development in AI-integrated preschool ecosystems.